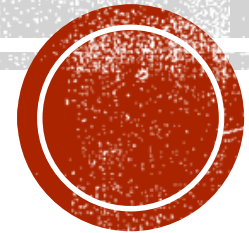




OBJECT-ORIENTED PROGRAMMING

IS2203



Lasanthi De Silva & Viraj Welgama

Lnc@ucsc.cmb.ac.lk; www@ucsc.cmb.ac.lk

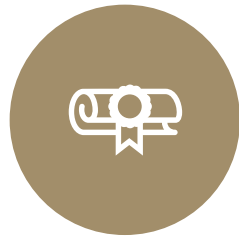
COURSE INTRODUCTION



COURSE SPECIFICS



NO OF CREDITS: 3
(2L + 1P)



SEMESTER: FIRST



CORE / OPTIONAL:
CORE



LECTURES:
2 HRS
(TOTAL: 30 HRS)



LAB SESSIONS:
2 HRS
(TOTAL: 30 HRS)

EVALUATION CRITERIA

- Assignments: 30%
 - Lab Session Submissions
 - Continuous in-class Assignments/ Take Home Assignment
- Final Exam: 70%

PASS MARK: 40



LECTURES & LAB SESSIONS

- Lectures: Thursdays from 10.00 am to 12.00 pm
- Lab Sessions
 - Day & Time: Tuesday 1 pm to 3 pm
- Course Notifications: via UGVLE

COURSE DESCRIPTION

This course introduces **Object-Oriented Programming (OOP)** using C++, covering key concepts such as **classes, objects, constructors/destructors, abstraction, encapsulation, inheritance, and polymorphism**. Students will also learn **templates** for generic programming and **exception handling** for robust error control.

Through hands-on projects, students will gain practical experience in writing efficient and maintainable code. The course builds a strong foundation for advanced programming and careers in software development.

INTENDED LEARNING OUTCOMES

By the end of this course, students will be able to:

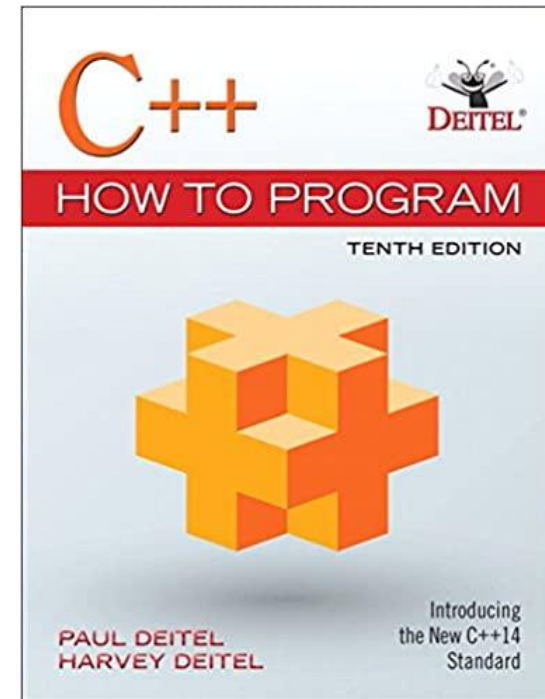
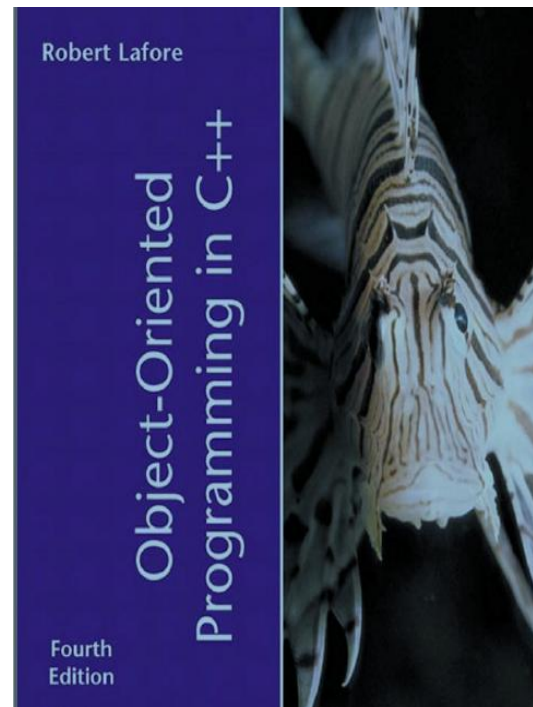
- Identify and apply key elements of classes in object-oriented programming.
- Explain and use constructors, destructors, function overloading, and operator overloading.
- Utilize reference variables, call-by-reference, default arguments, inline functions, and appropriate variable placement.
- Describe core OOP concepts such as data abstraction, encapsulation, inheritance, and polymorphism.
- Implement friend functions, friend classes, and virtual functions.
- Apply principles of single/multiple inheritance and aggregation.
- Design and implement generic programming solutions using templates.

COURSE CONTENT

- Introduction to object-oriented paradigm (1 Week)
- Constructors and Destructors (2 Weeks)
- Function Overloading (1 Week)
- Operator Overloading (2 Weeks)
- Data Conversion (1 Week)
- Inheritance (2 Weeks)
- Polymorphism (2 Weeks)
- Templates (1 Week)
- Exception Handling (1 Week)

COURSE REFERENCES

- Robert Lafore, 4th Edition, Object Oriented Programming in C++.
- Balagurusamy, E., 2021. Object-oriented programming with C++.





NEXT...

Introduction to Object Oriented Paradigm